CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **How do Organisms Reproduce?** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following correctly shows the flow of events in a human pregnancy? 1

(a) Fertilisation, implantation, placenta formation, birth
(b) birth, fertilisation, implantation
(c) implantation, fertilisation, birth
(d) placenta, birth, fertilisation

2. Assertion (A): Fragmentation is observed in Amoeba. Reason (R): Amoeba breaks into fragments, each becoming an individual. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The stage that is common to both sexual reproduction in humans and flowering plants is 1

(a) spore formation **(b)** fragmentation
(c) budding **(d)** fusion of male and female gametes

4. Why are the flowers of many plants brightly coloured and scented? 1

(a) To attract pollinators **(b)** To store food
(c) To keep cool **(d)** To prevent fertilisation

5. Assertion (A): Pollen tube is important for fertilisation. Reason (R): It carries male gametes to the ovule. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A girl's menstrual cycle is 28 days and menstruation begins on the 1st day. The egg is usually released about 1

(a) never **(b)** on day 1
(c) the middle of the cycle (around day 14) **(d)** on day 28

7. Which function is common to the testis and the ovary? 1

(a) Nourishing the embryo **(b)** Producing gametes and sex hormones
(c) Carrying gametes out of the body **(d)** Housing the foetus

8. Assertion (A): Sperms have tails. Reason (R): Tails help them swim to the egg. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. Which is correct about gametes? 1

(a) Human gametes have 23 chromosomes each **(b)** Sperm has 46 and egg 23

(c) Gametes are formed by fission **(d)** Human gametes have 46 chromosomes

10. The correct arrangement from outside to inside of a flower is 1

(a) carpels, stamens, petals, sepals **(b)** sepals, petals, stamens, carpels

(c) petals, sepals, carpels, stamens **(d)** stamens, petals, sepals, carpels

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Give the reason that a banana plant cannot be grown from seeds. 2

12. A flower is covered with a transparent bag before opening and no insects enter. It still forms fruit. What does this show? 2

13. A pregnant woman's menstruation stops. Why? 2

14. Why do the daughter cells formed by binary fission in a unicellular organism show very little variation? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student claims that regeneration and fragmentation are the same. Evaluate. 3

16. Sperms formed in the testis do not have the same number of chromosomes as the body cell. Explain how the number is restored. 3

17. Why is it advantageous for plants to have flowers with both male and female parts, and also for some to be unisexual? 3

OR

Describe an experiment to show that bread mould spreads through spores. 3

18. What are the roles of the prostate and seminal vesicles in reproduction? 3

OR

Compare the male and female gametes in terms of size, number and movement. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare a pea plant (bisexual flowers) and a papaya plant (unisexual flowers) in their modes of pollination, gamete production and the chances for variation. 5

OR

Trace the development of a flower into a fruit with seeds, naming the structure formed from each floral part. 5

20. Describe the male reproductive system and explain why sperm production would be affected if the testes were retained in the abdomen. 5

OR

Explain the changes in the uterus during one menstrual cycle in a woman who becomes pregnant and in one who does not. 5

END OF QUESTION PAPER — Total: 40 Marks